DIGITAL ELECTRONICS

ET 8301

Friday, November 22, 2024

Mr. Cuthbert J. Karawa

Lecture 05

THE KARNAUGH MAP

College of Information and Communication Technology (CoICT)

Department Of Electronics And Telecommunications Engineering.

Determining Standard Expressions from a Truth Table

- To determine the standard SOP expression represented by a truth table,
- List the binary values of the input variables for which the output is 1.
- Convert each binary value to the corresponding product term by replacing each 1 with the corresponding variable and each 0 with the corresponding variable complement.
- For example, the binary value 1010 is converted to a product term as $1010 \longrightarrow A\overline{B}C\overline{D}$

Determining Standard Expressions from a Truth Table

- To determine the standard POS expression represented by a truth table,
- List the binary values for which the output is 0.
- Convert each binary value to the corresponding sum term by replacing each 1 with the corresponding variable complement and each 0 with the corresponding variable.

$$1001 \longrightarrow \overline{A} + B + C + \overline{D}$$

Determining Standard Expressions from a Truth Table

- Example

 Determine the standard SOP expression and the equivalent standard POS expression

Inputs			Output
A	*B*	*C*	*X*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

SOP expression for the output X is

$$X = \overline{A}BC + A\overline{B}\,\overline{C} + AB\overline{C} + ABC$$

POS expression for the output X is

$$X = (A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + \overline{C})$$

The Karnaugh Map

- A Karnaugh map provides a systematic method for simplifying Boolean expressions and, if properly used, will produce the simplest **SOP** or **POS** expression possible, known as the ***minimum expression***.
- The effectiveness of ***algebraic simplification*** depends on your ***familiarity with all the laws***, ***rules***, and ***theorems of Boolean algebra*** and on your ability to apply them.

The Karnaugh Map

- The Karnaugh map its provides a "cookbook" method for simplification.
- Other simplification techniques include the ***Quine-McCluskey*** method and the ***Espresso algorithm***.
- A ***Karnaugh map*** is similar to a truth table because it presents all of the possible values of input variables and the resulting output for each value.
- The Karnaugh map is an ***array of cells*** in which each cell represents a binary value of the input variables.

The Karnaugh Map

- The cells are arranged in a way so that simplification of a given expression is simply a matter of properly grouping the cells.
- Karnaugh maps can be used for expressions with ***two***, ***three***, ***four***, and ***five*** variables.
- The ***number of cells*** in a Karnaugh map, as well as the number of rows in a truth table, is equal to the ***total number of possible input variable combinations***.
- For ***three variables***, the number of cells is **$2^3 = 8$**.
- For ***four variables***, the number of cells is **$2^4 = 16$**.

The Karnaugh Map

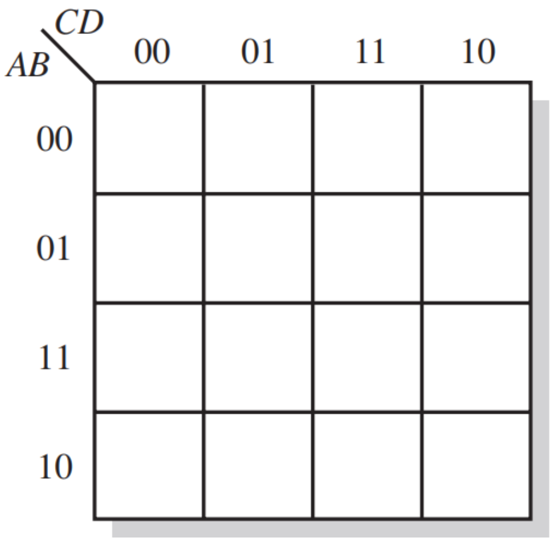
- The **3-Variable** Karnaugh Map
  - The 3-variable Karnaugh map is an array of ***eight*** cells.
  - Let use A, B, and C as variables.

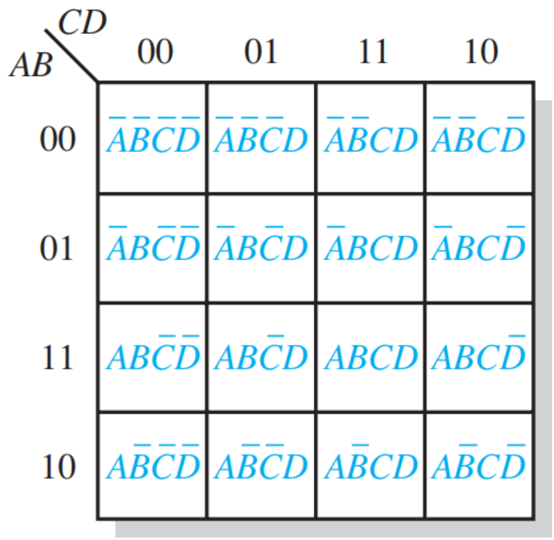
$AB \backslash C$	0	1
00		
01		
11		
10		

$AB \backslash C$	0	1
00	$\overline{A}\,\overline{B}\,\overline{C}$	$\overline{A}\,\overline{B}C$
01	$\overline{A}B\overline{C}$	$\overline{A}BC$
11	$AB\overline{C}$	ABC
10	$A\overline{B}\,\overline{C}$	$A\overline{B}C$

The Karnaugh Map

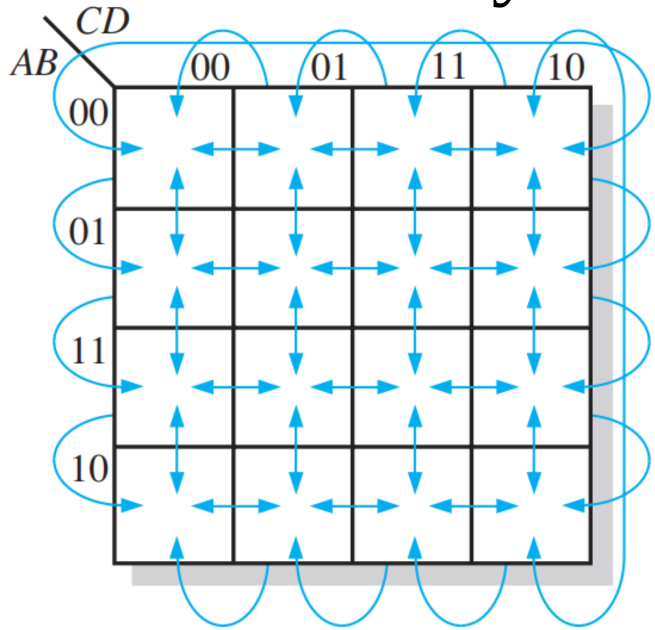
- The ***4-Variable*** Karnaugh Map.
  - The 4-variable Karnaugh map is an array of ***sixteen*** cells.





The Karnaugh Map

- Adjacency is defined by a ***single-variable change***.
- For example in the 3-variable map the **010** cell is adjacent to the **000** cell, the **011** cell, and the **110** cell.
- Physically, each cell is adjacent to the cells that are immediately next to it on any of its four sides.



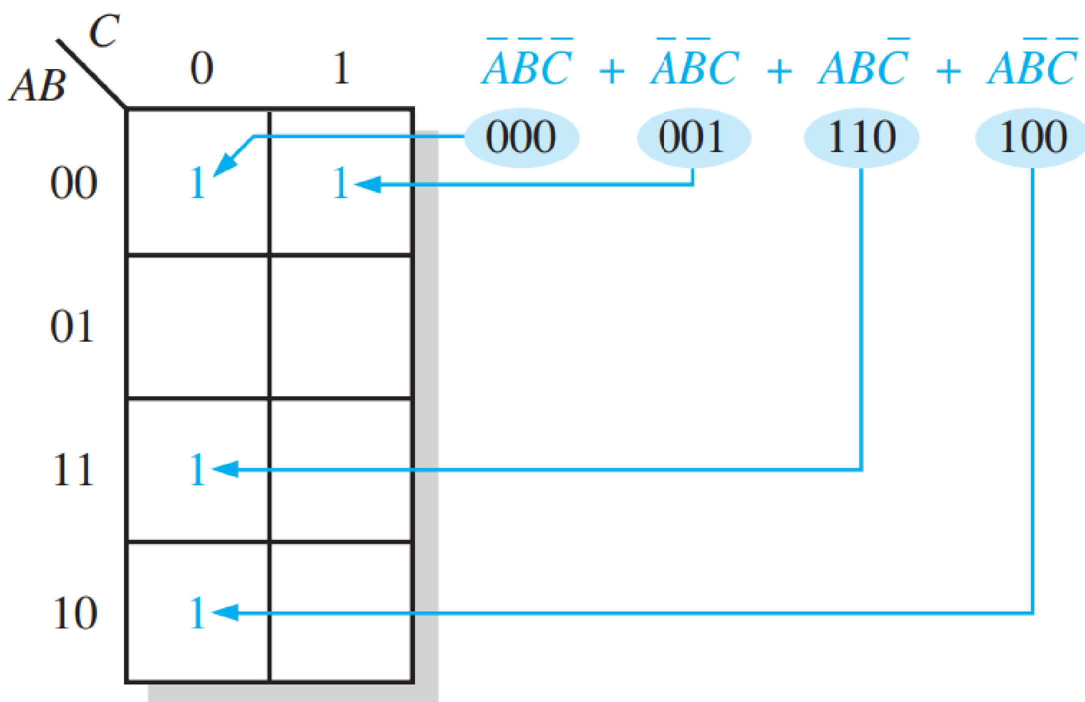
The Karnaugh Map

SOP Minimization

- A minimum SOP expression can be implemented with fewer ***logic gates*** than a standard expression.
- For an SOP expression in standard form, a **1** is placed on the Karnaugh map for each product term in the expression.
- Each **1** is placed in a **cell** corresponding to the value of a product term.
- Number of **1s** on the Karnaugh map ***equal*** to the number of product terms in the standard SOP expression.

The Karnaugh Map

- Mapping a standard SOP expression



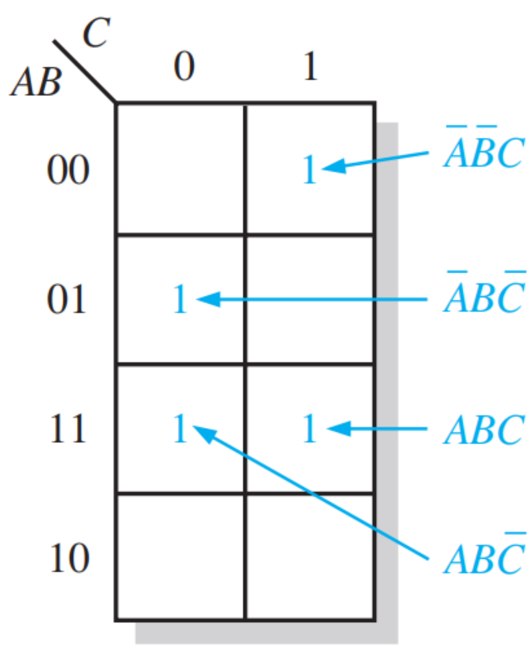
The Karnaugh Map

- Example

  Map the following standard SOP expression on a Karnaugh map

$$\overline{A}\,\overline{B}C + \overline{A}B\overline{C} + AB\overline{C} + ABC$$

Solution

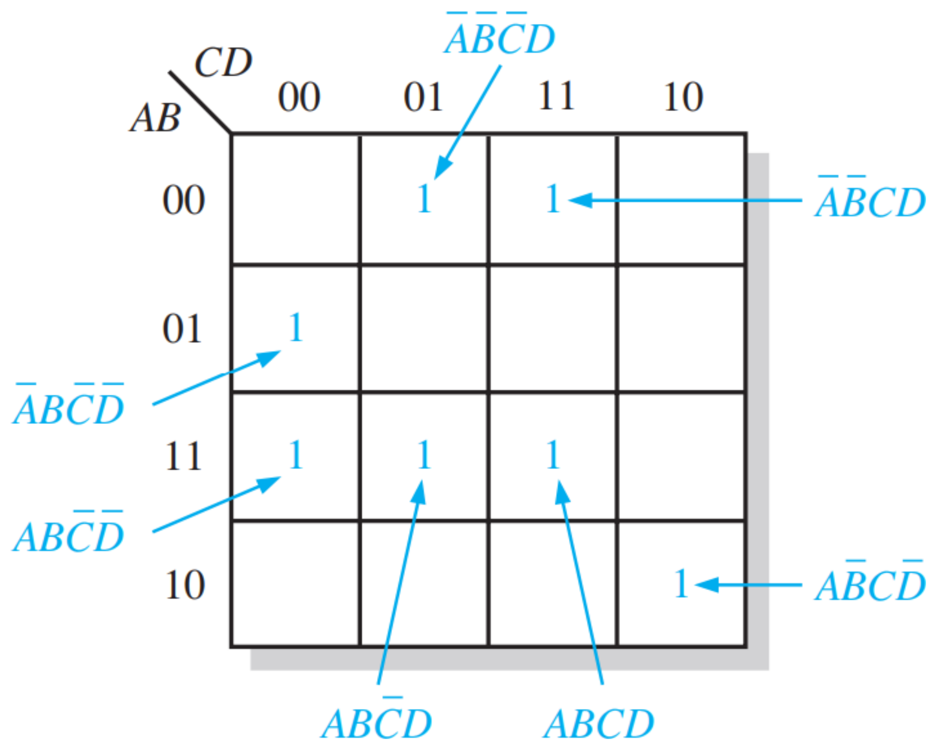


The Karnaugh Map

Example

Map the following standard SOP expression on a Karnaugh map: $\overline{A}\,\overline{B}CD + \overline{A}B\overline{C}\,\overline{D} + AB\overline{C}D + ABCD + AB\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,\overline{C}D + A\overline{B}C\overline{D}$

Solution



The Karnaugh Map

- A Boolean expression must first be in standard form before you use a Karnaugh map.
- If an expression is not in standard form, then it must be converted to standard form.
- Nonstandard product term has ***one or more missing*** variables.
- For example, assume that one of the product terms in a certain 3-variable SOP expression is AB.

The Karnaugh Map

Example : Map the following SOP expression on a Karnaugh map:

$$\overline{A} + A\overline{B} + AB\overline{C}.$$

- Solution

$\overline{A}$	$+ A\overline{B}$	$+ AB\overline{C}$
000	100	110
001	101	
010		
011		

$AB \backslash C$	0	1
00	1	1
01	1	1
11	1	
10	1	1

The Karnaugh Map

Minimization

- After an SOP expression has been mapped, a minimum SOP expression is obtained ***by grouping the 1s*** and determining the minimum SOP expression from the map.
- You can ***group 1s*** on the Karnaugh map according to the rules ***by enclosing those adjacent cells containing 1s***.
- The goal is to ***maximize the size of the groups*** and ***to minimize the number of groups***.

The Karnaugh Map

Rules

1. A group must contain either 1, 2, 4, 8, or 16 cells, which are all powers of two. In the case of a 3-variable map, 23 = 8 cells is the maximum group.
2. Each cell in a group must be adjacent to one or more cells in that same group, but all cells in the group do not have to be adjacent to each other.
3. Always include the largest possible number of 1s in a group in accordance with rule 1.
4. Each 1 on the map must be included in at least one group. The 1s already in a group can be included in another group as long as the overlapping groups include noncommon 1s.

The Karnaugh Map

Example:

Group the 1s in each of the Karnaugh maps

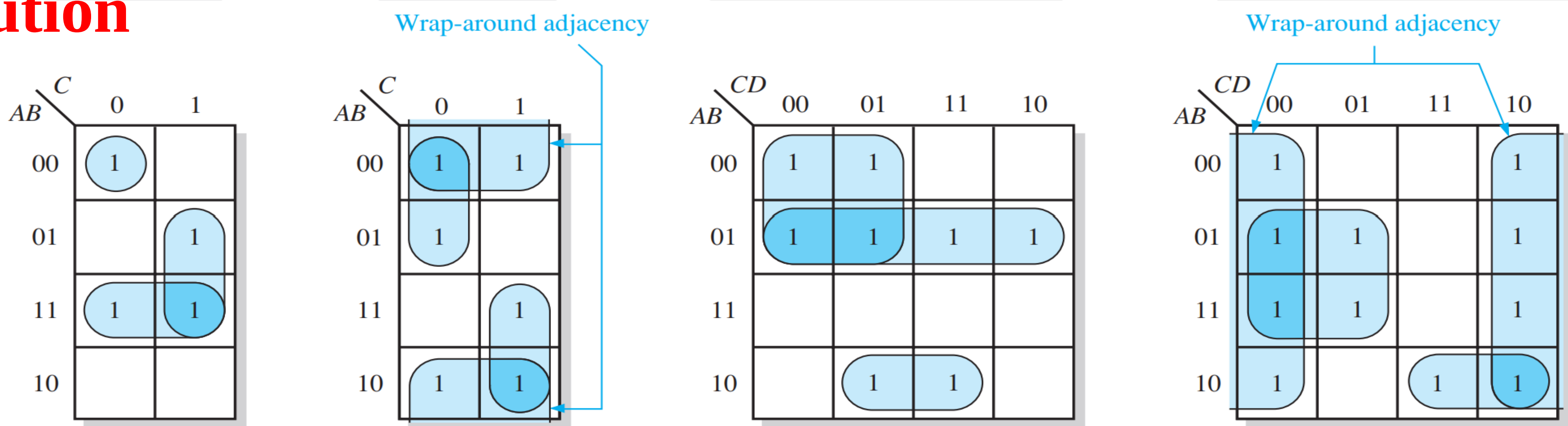
AB \ C	0	1
00	1	
01		1
11	1	1
10		

AB \ C	0	1
00	1	1
01	1	
11		1
10	1	1

AB \ CD	00	01	11	10
00	1	1		
01	1	1	1	1
11				
10		1	1	

AB \ CD	00	01	11	10
00	1			1
01	1	1		1
11	1	1		1
10	1		1	1

Solution



Next Lecture

QUINE- McClUSKEY METHOD